Practical No – 5

Perform the linear regression on the given data warehouse data using R/Python.

Input data below is the sample data representing the observations:

Values of height 151, 174, 138, 186, 128, 136, 179, 163, 152, 131

Values of weight. 63, 81, 56, 91, 47, 57, 76, 72, 62, 48

lm () function:-

This function creates the relationship model between the predictor and the response variable

Syntax: - lm (formula, data)

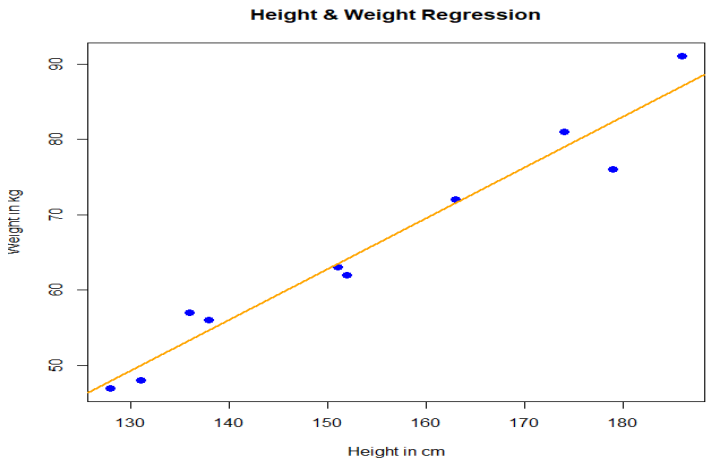
Following is the description of the parameters used: −

• Formula is a symbol presenting the relation between x and y.

• Data is the vector on which the formula will be applied.

CODE:-

```
x<-c (151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y<-c (63, 81, 56, 91, 47, 57, 76, 72, 62, 48)
relation<-lm(y~x) # Create the linear regression model.
print(relation) # Print the relationship model.
summary(relation)
new_height<-data.frame(x=190)
predict(relation,new_height)
# Plot the chart (the plot is saved to the file, not displayed in RStudio)
plot(x,y,
    col = "blue",
    main = "Height & Weight Regression",
    xlab = "Height in cm",
    ylab = "Weight in kg",
    pch = 16,
    cex = 1.3)
abline(relation,col="orange",lwd=2) # Add regression line
```



Practical No – 6

Perform the logistic regression on the given data warehouse data using R/Python.

```
data(mtcars)
input<-mtcars[, c("am", "cyl", "hp", "wt")]
print(head(input))
```

```
                   am cyl  hp    wt
Mazda RX4           1   6 110 2.620
Mazda RX4 Wag       1   6 110 2.875
Datsun 710          1   4  93 2.320
Hornet 4 Drive      0   6 110 3.215
Hornet Sportabout   0   8 175 3.440
Valiant             0   6 105 3.460
```

```
am.data <- glm(
  am ~ cyl + hp + wt,
  data = input,
  family = binomial,
  control = glm.control(trace = TRUE)
)
summary(am.data)
```

Output:

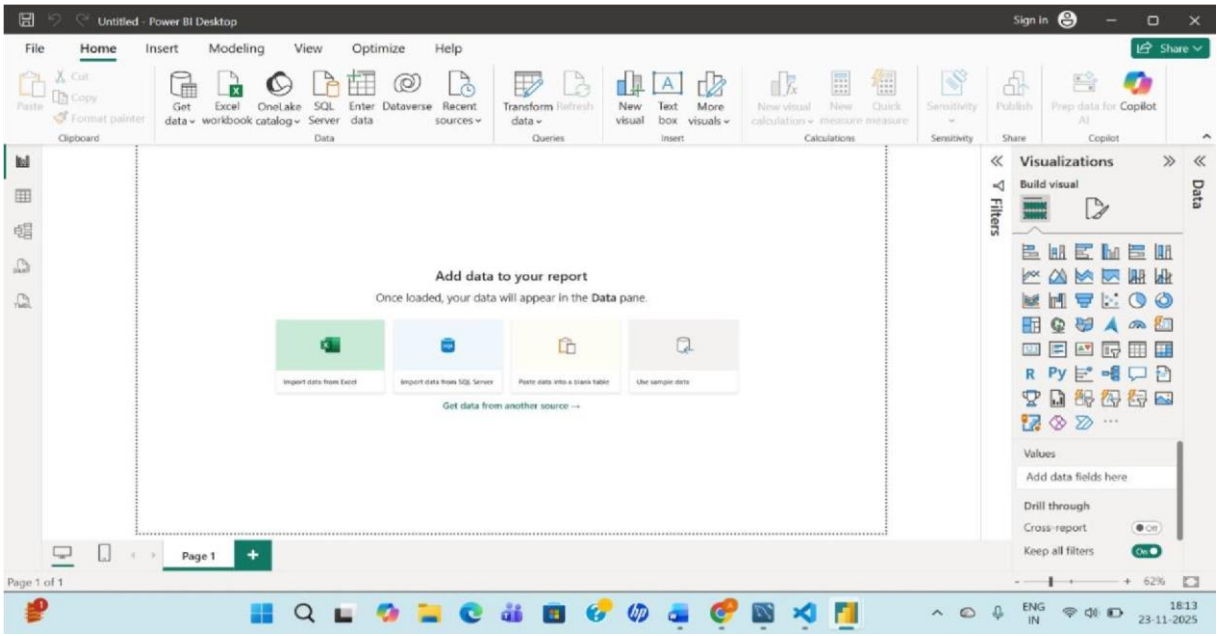
```
Call:
glm(formula = am ~ cyl + hp + wt, family = binomial, data = input,
    control = glm.control(trace = TRUE))

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 19.70288    8.11637   2.428   0.0152 *
cyl          0.48760    1.07162   0.455   0.6491
hp           0.03259    0.01886   1.728   0.0840 .
wt          -9.14947    4.15332  -2.203   0.0276 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 43.2297  on 31  degrees of freedom
Residual deviance:  9.8415  on 28  degrees of freedom
AIC: 17.841

Number of Fisher Scoring iterations: 8
```

Practical No – 8

B. Perform data visualization using Power BI on any sales data.

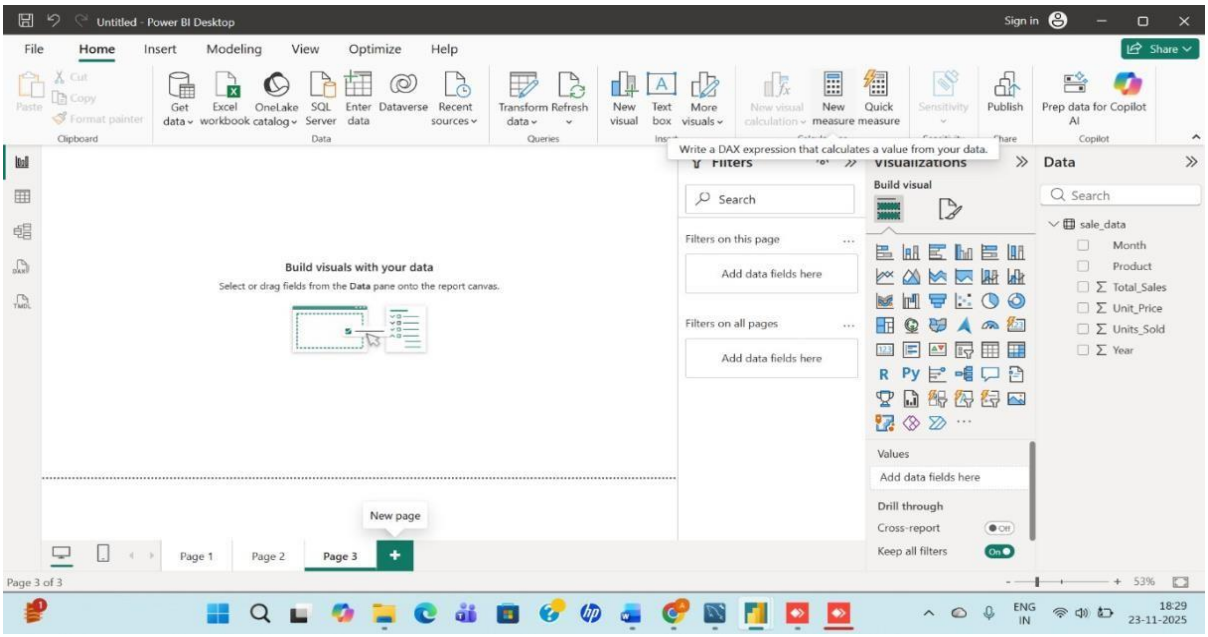
Step 1: Open power BI Desktop

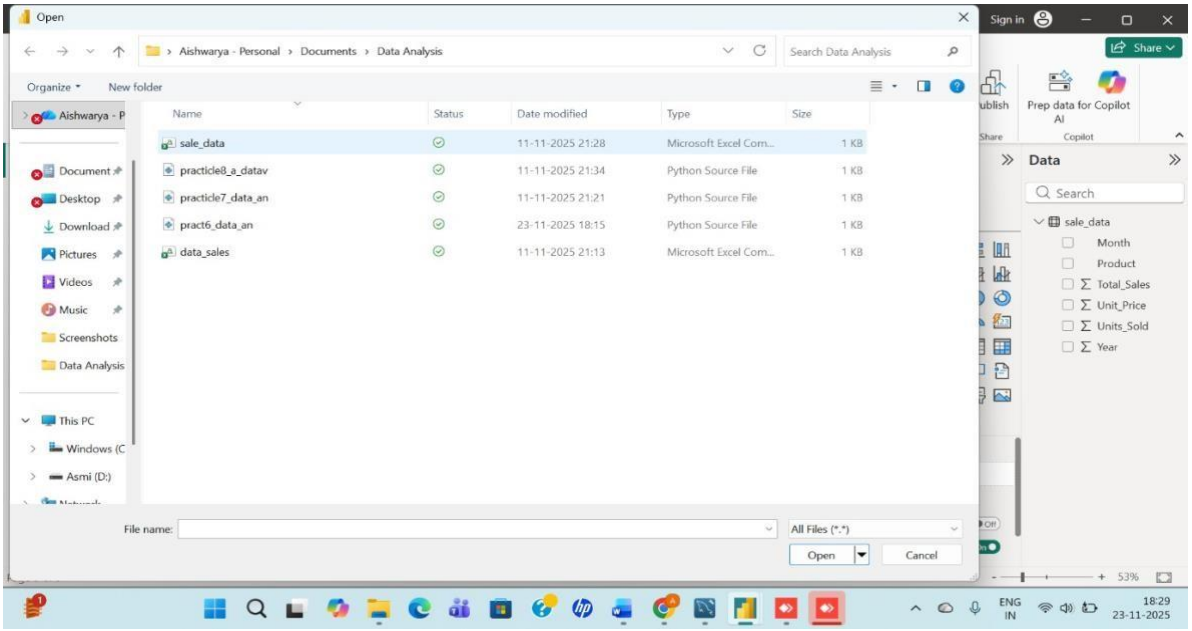
- Launch **Power BI Desktop.**
- You will see a blank report canvas with options to import data.



Step 2: Import Data into Power BI

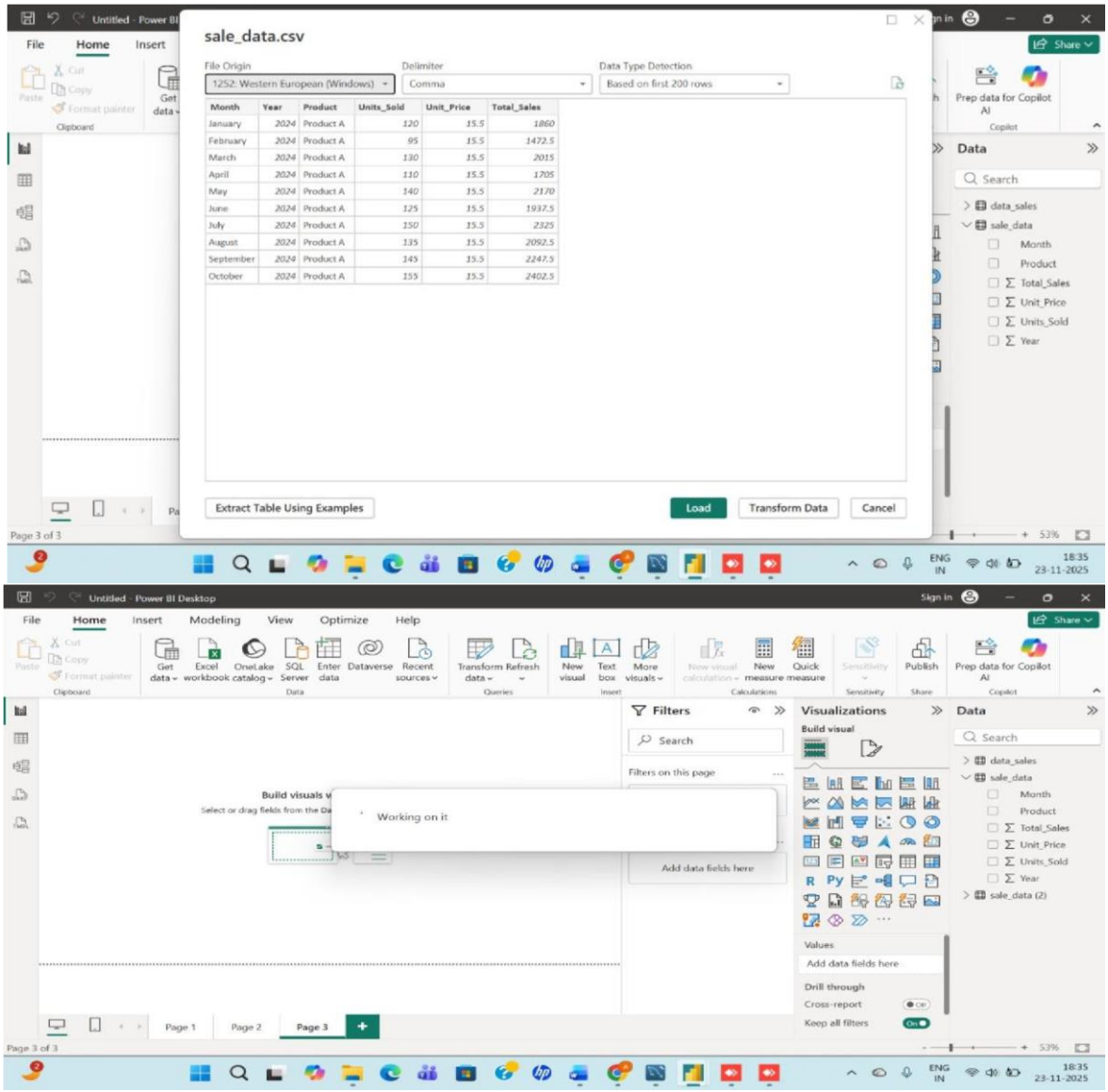
1. Go to **Home → Get Data.**
2. Choose **Excel** or **Text/CSV** depending on your file.
3. Select file **sale_data.csv** from your computer.
4. Preview windows opens showing all columns (Month, Year, Product, Units_Sold, Unit_Price, Total_Sales).
5. Click **Load** to import the dataset into Power BI.





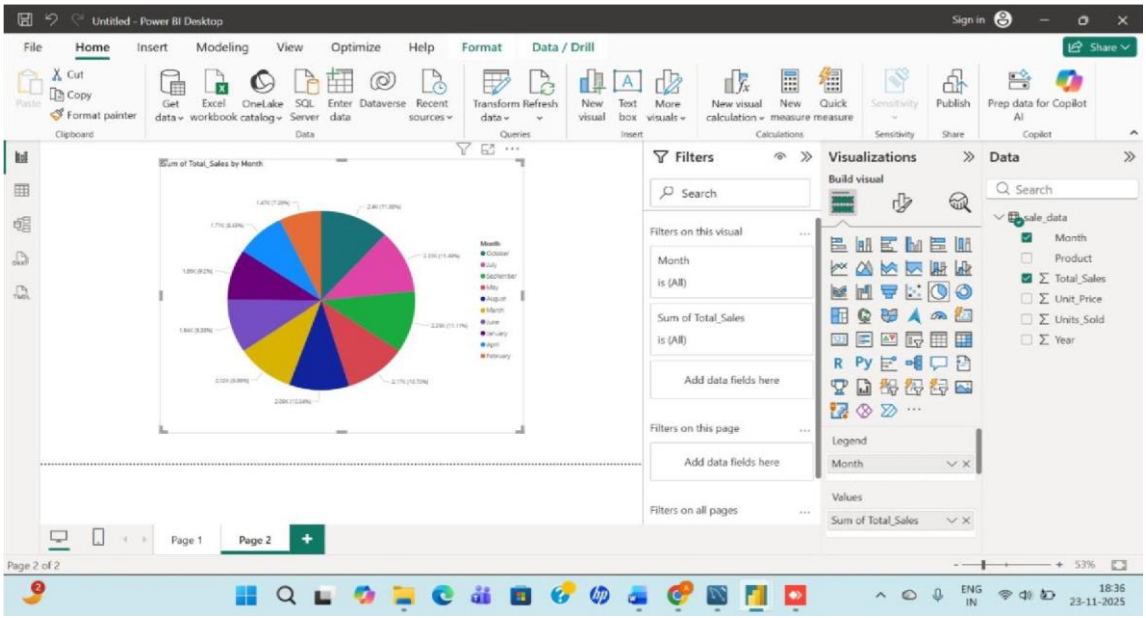
Step 3: View the Imported Dataset

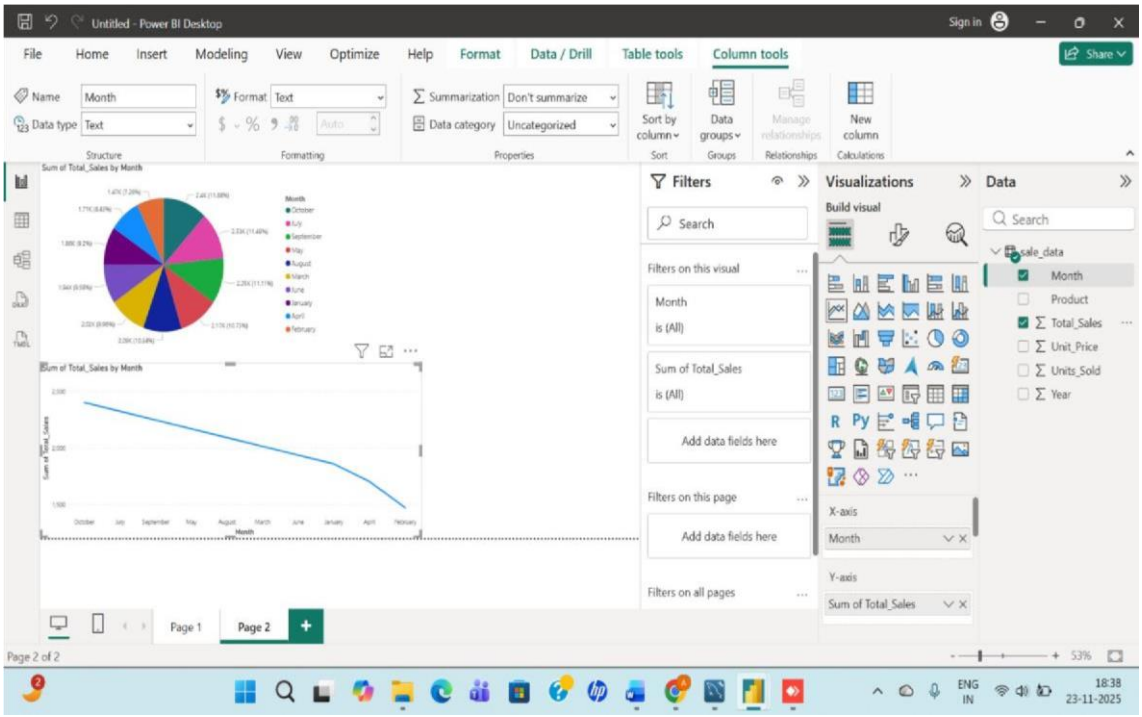
- After loading, the dataset **sale_data** appears in the **Data pane** on the right side.
- Columns such as Month, Total_Sales, Units_Sold, etc. become available for visualization.



Step 4: Create a Pie Chart of Monthly Total Sales

1. Go to **visualization pane** and click the **Pie Chart** icon.
2. A blank chart appears on the canvas.
3. Set the fields:
   - **Legend:** Month
   - **Values:** Total_Sales
4. The pie chart will display the percentage share of each month's total sales.



Step 5: Create a Line Chart of Total Sales by Month

1. Click on the **Line Chart** icon in the Visualizations pane.
2. Drag fields into these areas:
   - **X-axis:** Month
   - **Y-axis:** Total_Sales
3. The line chart displays month-wise sales trend.